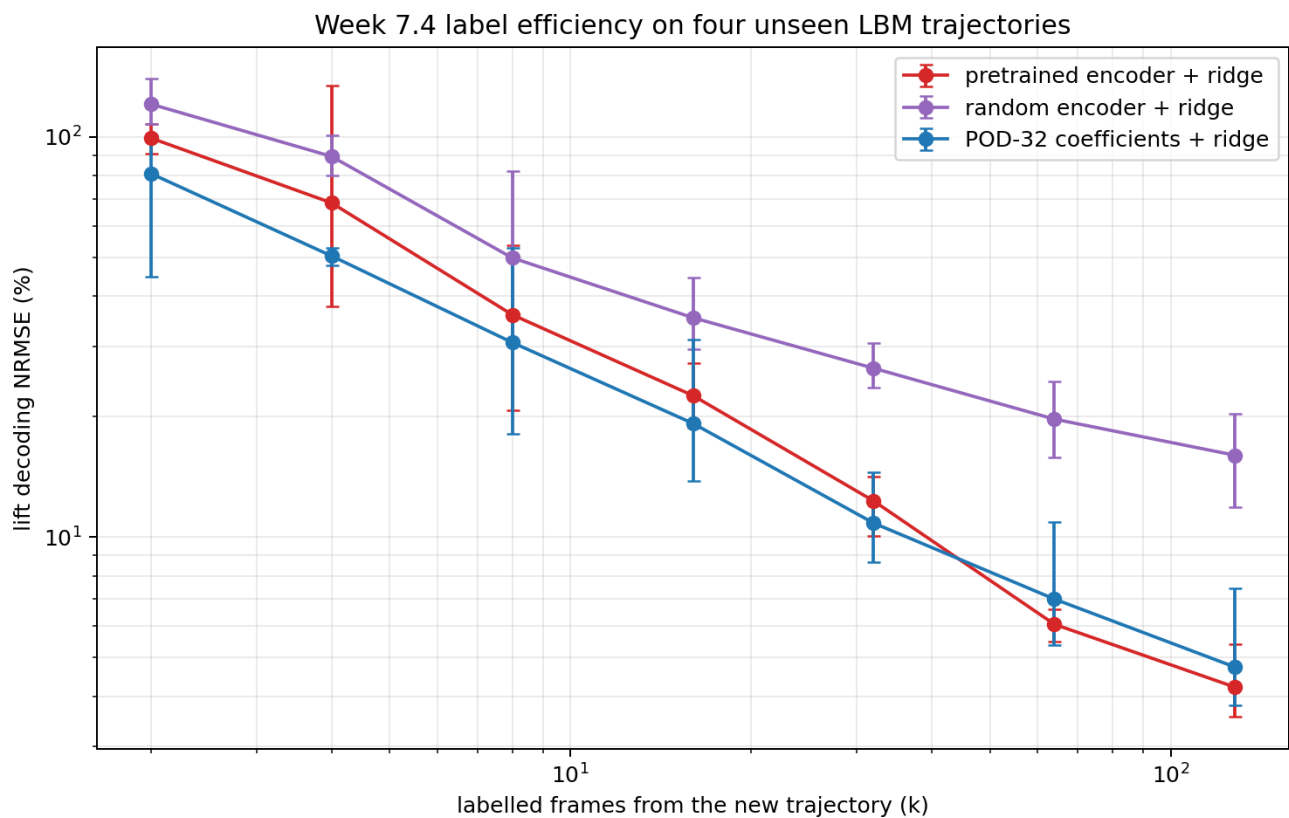


## Week 7.4

# Diverse-wake pretraining and label-efficient force decoding

A trajectory-split, MAPA-inspired experiment on 16 educational LBM wakes



## Why a new week?

Week 7.3 asks a network to complete a periodic low-rank field, a task gappy POD almost solves analytically. MAPA transfers a representation learned from heterogeneous unlabelled signals to an external label. Week 7.4 follows that design: the MAE sees velocity only; the downstream target is cylinder lift.

## Frozen split

Development: Re 60, 65, 70, 75, 80, 85, 90, 100, 110, 120, 130. Validation: Re 105. Untouched test: Re 95, 115, 125, 135. Target labels come from frames [0,150); frames [180,251) are scored. No random frame split crosses trajectory roles.

## Model

A 90k-parameter masked autoencoder hides 75% of 4 by 6 patches and trains for 6000 AdamW steps. The encoder sees visible tokens only. Downstream, all visible tokens are mean pooled to a 64-component representation; ridge strength is selected on Re105.

## Baselines

The identical random encoder tests the value of pretraining. POD-32 coefficients provide a strong classical representation. Every representation receives exactly the same k labels and ridge protocol.

## Definitions and reading the metric

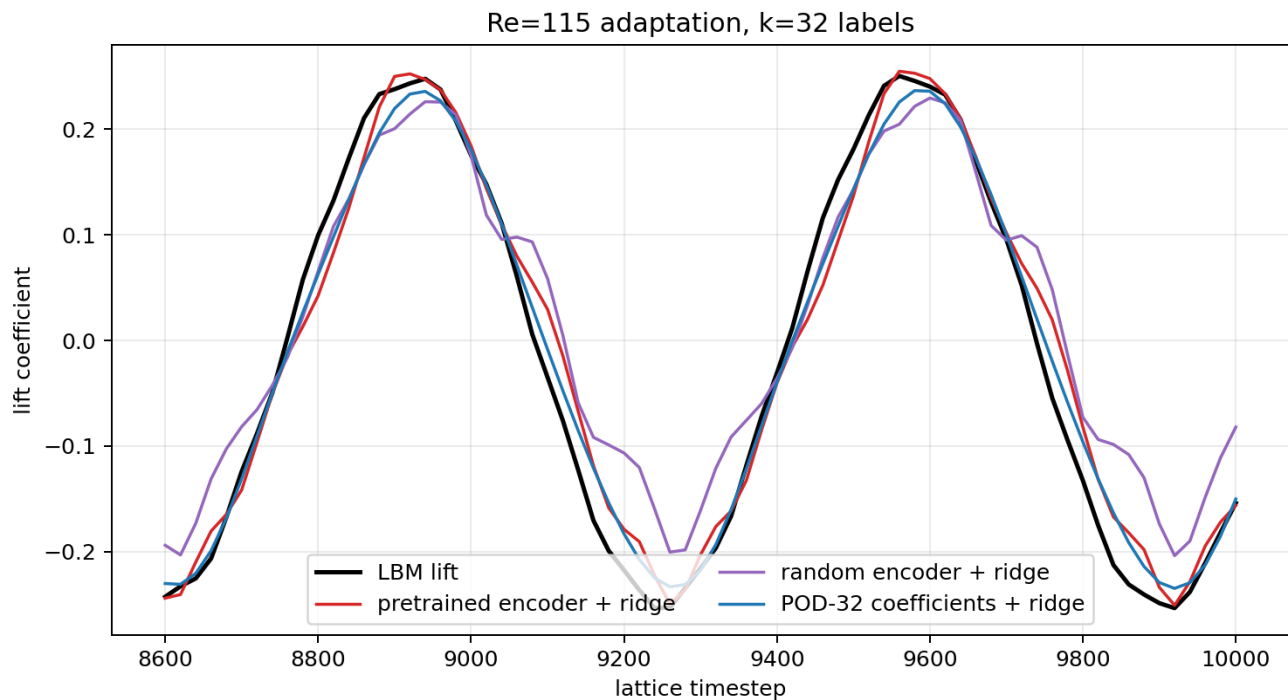
A 32 by 78 snapshot is divided into 104 patches, each with 24 velocity values. The hidden-patch MSE supplies the pretraining signal. The downstream ridge readout minimizes squared lift error plus an L2 penalty on its weights. NRMSE (%) is 100 times RMSE divided by the scored lift standard deviation. The current field is available when decoding current lift; this is not an autonomous forecast. Lift is produced alongside the solver fields, so label scarcity is simulated rather than an observed saving in CFD cost.

## Retained numerical evidence

Lift NRMSE (%), mean across four unseen trajectories

| k   | Pretrained | Random | POD-32 |
|-----|------------|--------|--------|
| 2   | 99.44      | 120.91 | 81.06  |
| 4   | 68.46      | 89.45  | 50.44  |
| 8   | 35.96      | 49.89  | 30.65  |
| 16  | 22.59      | 35.35  | 19.24  |
| 32  | 12.31      | 26.43  | 10.84  |
| 64  | 6.05       | 19.72  | 7.00   |
| 128 | 4.22       | 16.02  | 4.73   |

At  $k=32$ , pretrained 12.31% is below random-at- $k=128$  16.02%: better mean accuracy with one-quarter as many target labels. POD leads through  $k=32$ ; pretrained passes it in the mean at  $k=64$  and 128.



## Scope and reproducibility

Ridge selection uses 2761 development and 251 validation lift labels in addition to the reported target-label budget. Only one encoder initialization is retained; the three seeds are label draws. POD rank 32 is fixed, not rank-optimized. One run per Reynolds number confounds Reynolds and initial-condition variation. The test cases are now inspected retained evidence; future tuning needs fresh tests. Mean differences are not statistical significance claims.

The retained success is representation transfer for an external force label, not a claim that transformers universally beat POD. The solver uses  $D=6$  quick D2Q9-TRT trajectories and periodic transverse boundaries; the data are not grid-independent DNS. Re140 was rejected after the solver became non-physical. The histories do not support a validated Strouhal claim.

Reproduce: generate data with `qa/generate_week07_4_wakes.py`; train and score with `qa/run_week07_4_protocol.py`; inspect hashes and every record in `results/week07_4_diverse_pretraining/metrics.json`.

## References

Tang, Spalding and Cogan (2026), Pretraining for Sample-Efficient Neural Interfaces (MAPA), arXiv:2609.13507. He et al. (2022), Masked Autoencoders Are Scalable Vision Learners. Everson and Sirovich (1995), Karhunen-Loeve procedure for gappy data.